



Synthetic and scalable data platform for medical empowered AI



Funded by the European Union under Grant Agreement N.210810650. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the granting authorities. Neither the European Union nor the granting authorities can be held responsible for them.



Project funded by

 Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research EAER
State Secretariat for Education,
Research and Innovation SERI



AISYN4MED

Aim: Facilitate the development of robust AI solutions in healthcare by leveraging synthetic data generation and auditing, ensuring secure data sharing to improve medical research and patient outcomes.

Consortium

- *Strong network of partners*
- *Interdisciplinary*
- *Different backgrounds*
- *Complementary knowledge, skills and resources.*



15 partners

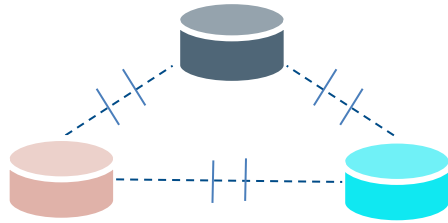
8 countries

4 years

+6M€
Total budget

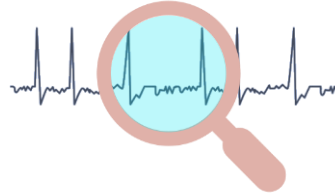
Motivation

The basis of AI in medicine is healthcare data, which is:



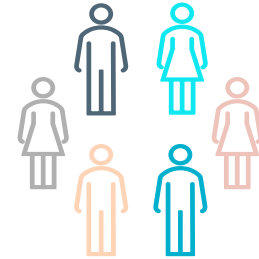
1

- *Highly sensitive*
- *Private*
- *Scattered or Siloed*



2

- *Incomplete*
- *Imbalanced*
- *Unstructured*



3

- *Biased*
- *Underrepresenting minority populations and rare diseases*

Regulatory transition in Digital Health



BRIEFING
EU Legislation in Progress



Artificial intelligence act

Medical Device Regulation: Software as a medical device

REGULATION (EU) 2017/745 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL
of 5 April 2017

on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing
Council Directives 90/385/EEC and 93/42/EEC

Needs

Healthcare Data



1

*Privacy-preservation
Harmonisation
Standardisation
Improved quality and completeness*

AI-Based Solutions



2

Access to better data:

- *Large amounts*
- *High-quality*
- *Representative*

Real-world validation & auditing

Improved Care

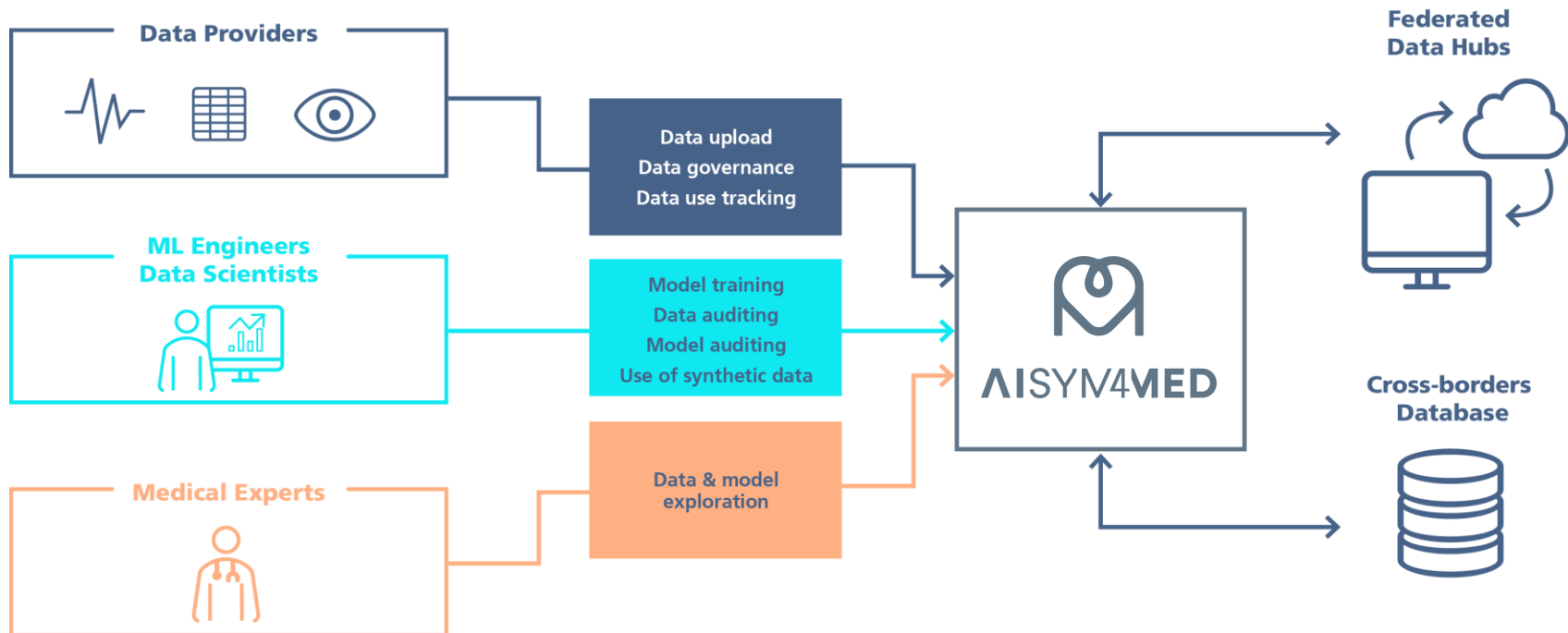


3

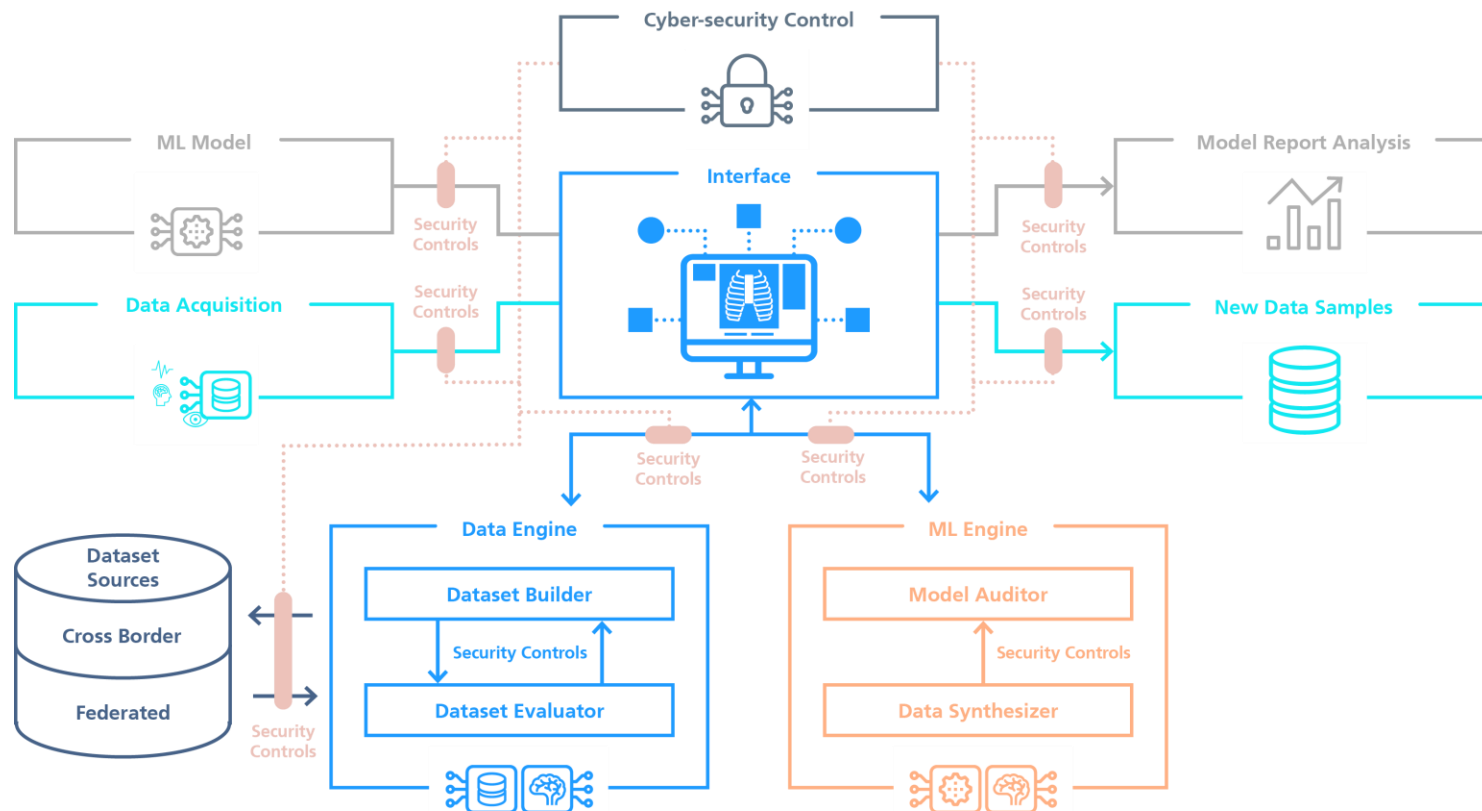
*Reduction of bias
Increased efficiency
Knowledge sharing
Consistency
Better representation of rare pathologies*

AlSym4Med
Platform

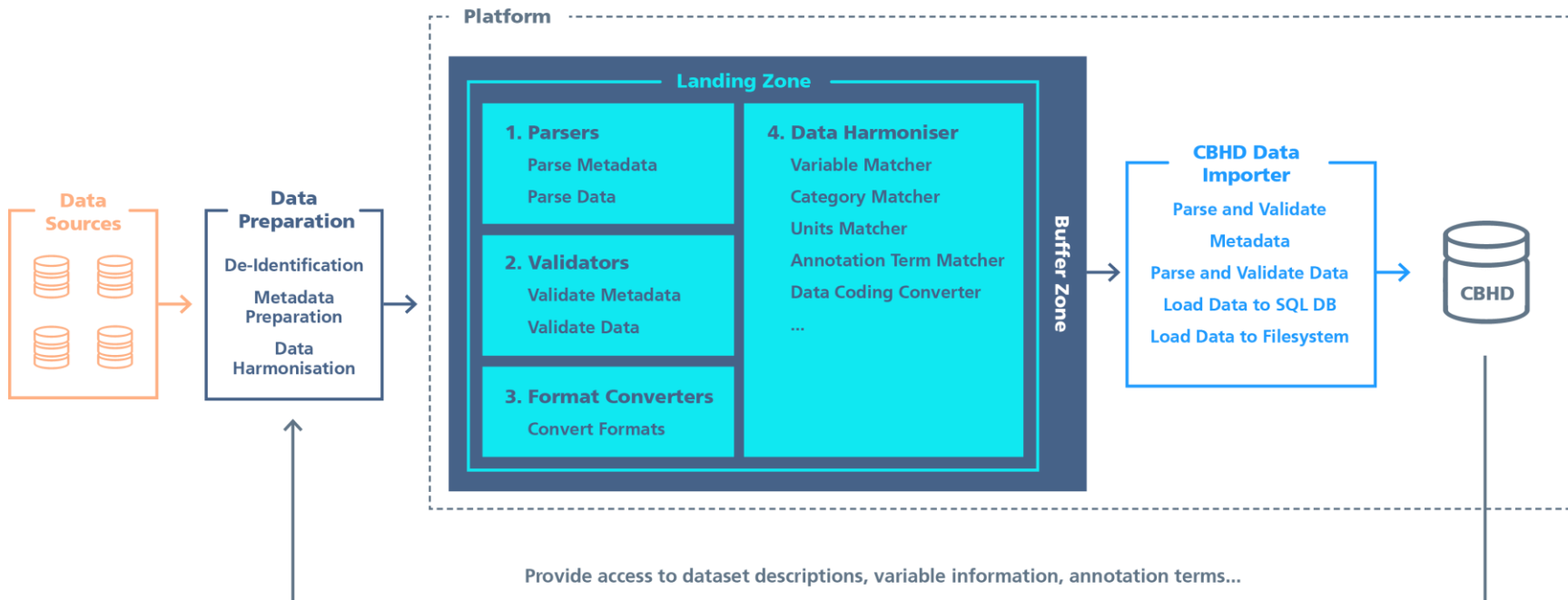
AI Sym4Med Concept



Architecture Design



Data Ingestion Process





AISYN4MED

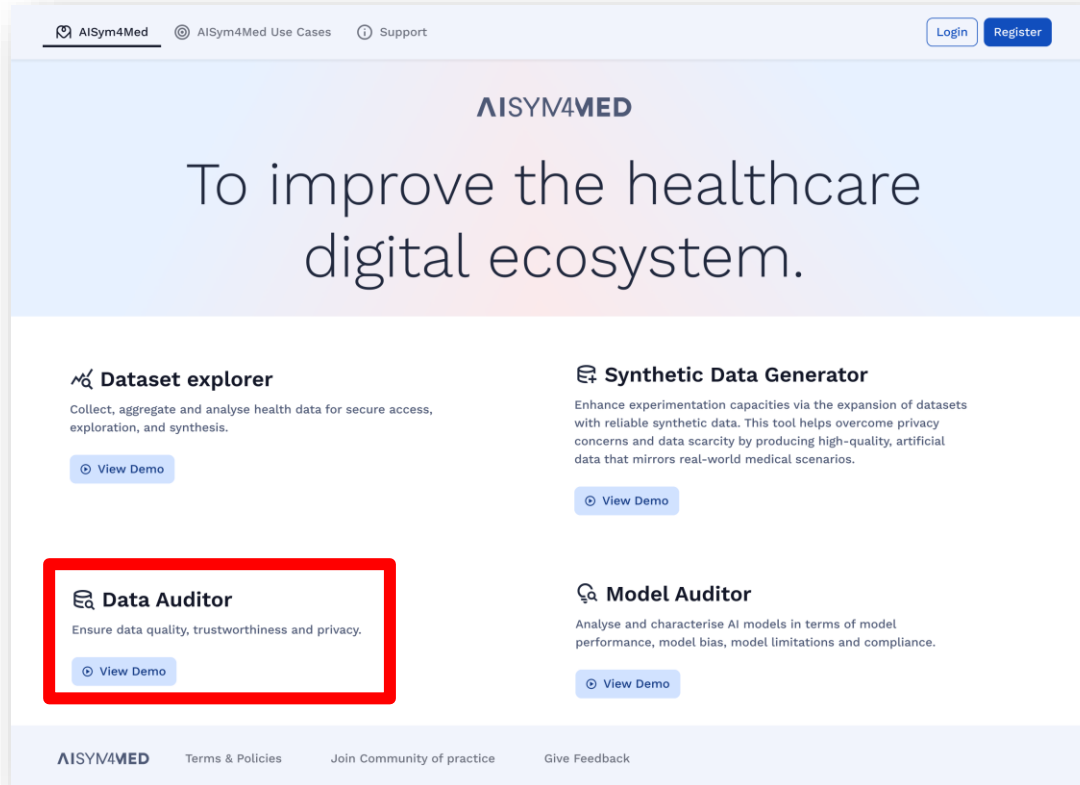


**UK Research
and Innovation**



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

AI Sym4Med Modules



pyMDMA: Multimodal Data Metrics for Auditing Real and Synthetic Data

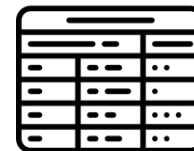
1

Open-source **Python** library with standardized metric implementations based on **NumPy** and **PyTorch**



2

Comprehensive suite of metrics for assessing **real** and **synthetic** datasets as **easily** and **fast** as possible



3

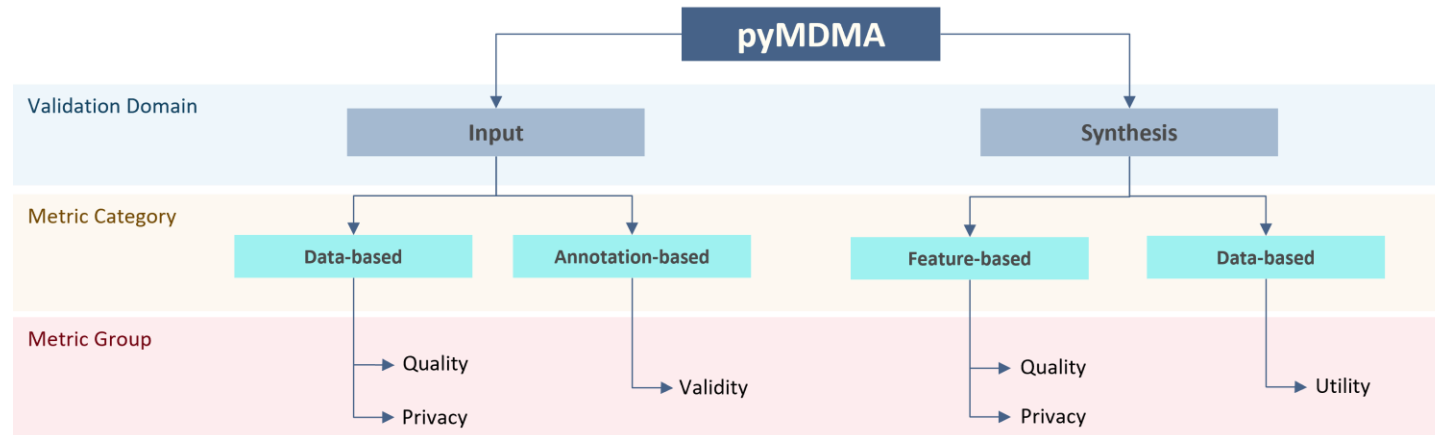
Metrics for **Image**, **Tabular** and **Time-Series** Data Modalities categorized on **evaluation type** and **requirements**



pyMDMA

Metric Hierarchy

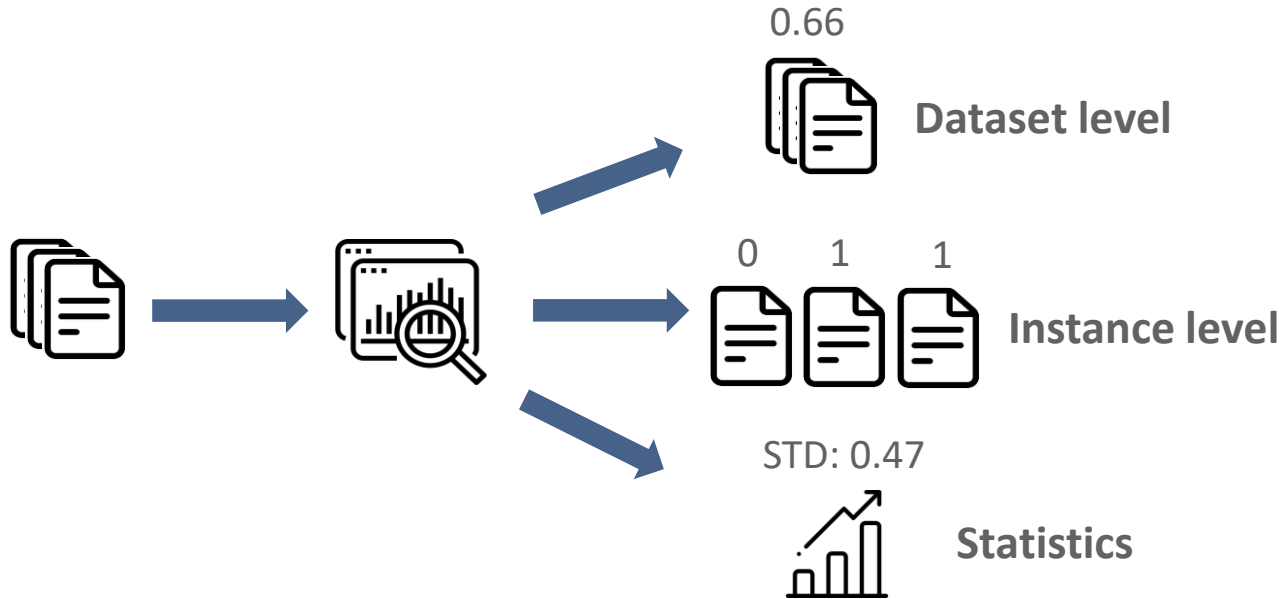
- Metrics are organized on data **characteristics** and **computation requirements**
- **Validation domain** is related to the **type** of the data
- **Metric category** groups metrics by **type of data** they consume
- **Metric group** defines the general metric **purpose**

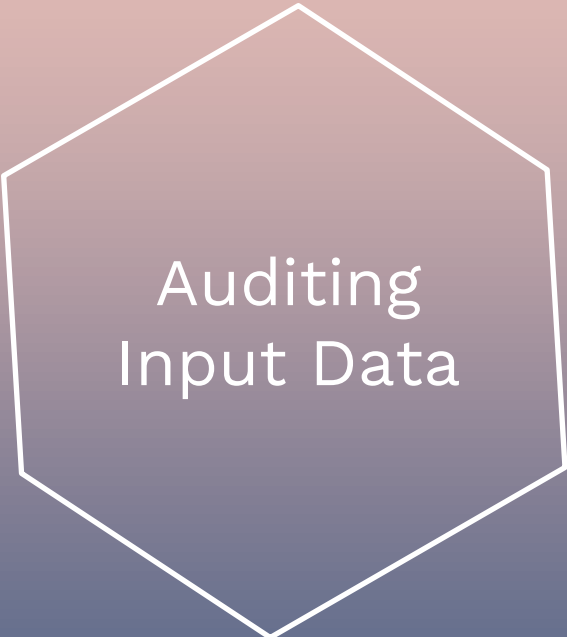


pyMDMA

Metric Output Class

- Standardized metric **output class** to enforce return types
- Metrics can evaluate single **instances** or the **dataset** as a whole





Auditing Input Data

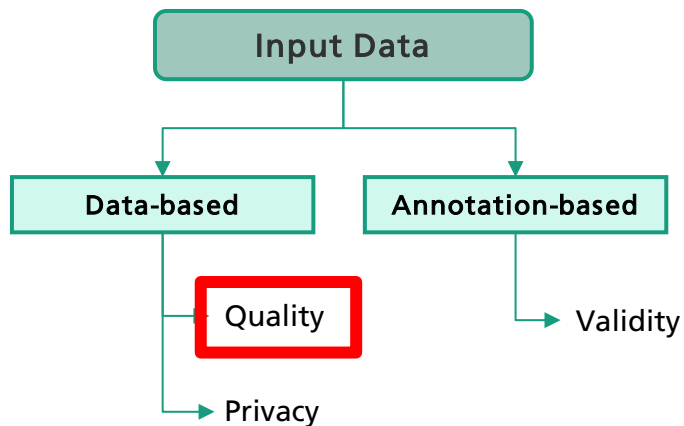
pyMDMA

Input Data Auditing Metrics

Modality	Metric	Output	Ref. Type	Category	Group	Goal	Reference
Image	DOM	Instance	None	Data-based	Quality	Sharpness	Kumar et al. (2012)
	Tenengrad	Instance	None	Data-based	Quality	Sharpness	Groen et al. (1985)
	TenengradRelative	Instance	None	Data-based	Quality	Sharpness	Rosado et al. (2018)
	EME	Instance	None	Data-based	Quality	Contrast	Panetta and Grigoryan (2000)
	ExposureBrightness	Instance	None	Data-based	Quality	Brightness	—
	Brightness	Instance	None	Data-based	Quality	Brightness	Finley (2006)
	Colorfulness	Instance	None	Data-based	Quality	Colorfulness	Rajashekar et al. (2009)
	CLIPQA	Instance	None	Data-based	Quality	Perceptual Quality	Wang et al. (2023)
	BRISQUE	Instance	None	Data-based	Quality	Perceptual Quality	Mittal et al. (2012)
	PSNR	Instance	Instance	Data-based	Quality	Signal to Noise Ratio	Gonzalez and Woods (2018)
	SSIM	Instance	Instance	Data-based	Quality	Structural Similarity	Wang et al. (2004)
	MS-SSIM	Instance	Instance	Data-based	Quality	Structural Similarity	Wang et al. (2003)
	DatasetCompleteness	Dataset, Instance	None	Annotation-based	Validity	Completeness	Lin et al. (2015)
Time Series	AnnotationCorrectness	Dataset, Instance	None	Annotation-based	Validity	Correctness	Lin et al. (2015)
	AnnotationUniqueness	Dataset, Instance	None	Annotation-based	Validity	Uniqueness	Lin et al. (2015)
Time Series	SNR	Dataset	None	Data-based	Quality	Signal to Noise Ratio	Smith (1997)
	Uniqueness	Dataset	None	Data-based	Quality	Uniqueness	—
Tabular	CorrelationScore	Dataset	None	Data-based	Quality	Correlation	Shrestha (2020)
	UniquenessScore	Dataset	None	Data-based	Quality	Uniqueness	Sukhobok et al. (2017)
	UniformityScore	Dataset	None	Data-based	Quality	Uniformity	—
	OutlierScore	Dataset	None	Data-based	Quality	Outlier Detection	Iglewicz and Hoaglin (1993)
	MissingScore	Dataset	None	Data-based	Quality	Completeness	Taleb et al. (2016)
	DimCurseScore	Dataset	None	Data-based	Quality	Dimensionality	—
	VIFactorScore	Dataset	None	Data-based	Quality	Multicollinearity	Marcoulides and Raykov (2019)
	KAnonymityScore	Dataset	None	Data-based	Privacy	Privacy	Díaz and García (2022)

pyMDMA

Input Data Auditing Metrics



Dataset-level Image Quality Assessment (IQA)

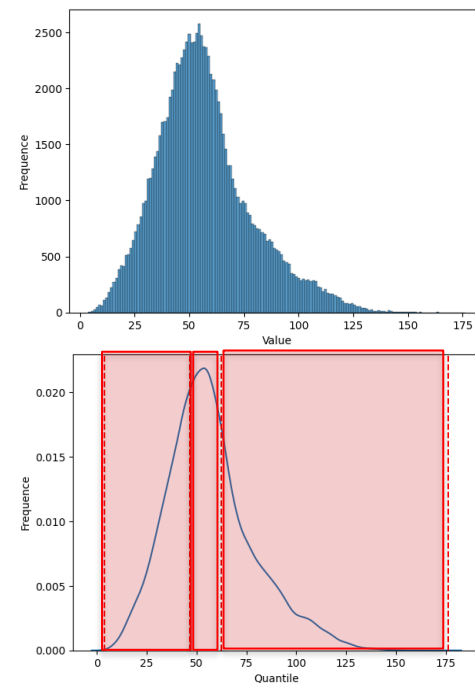
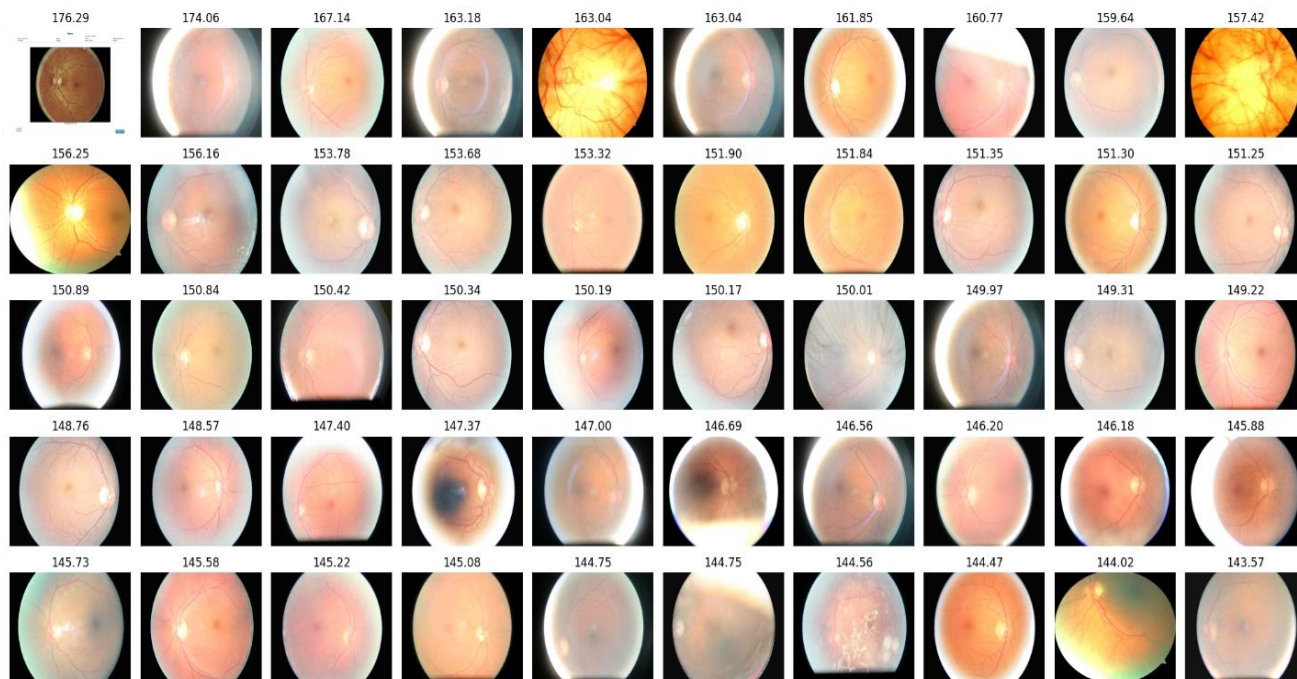
Comprehensive suite of metrics for quality assessment of input image data:

- Contrast
- Brightness
- Perceptual quality
- Colourfulness
- Noise
- ...

pyMDMA

Input Data Auditing Metrics

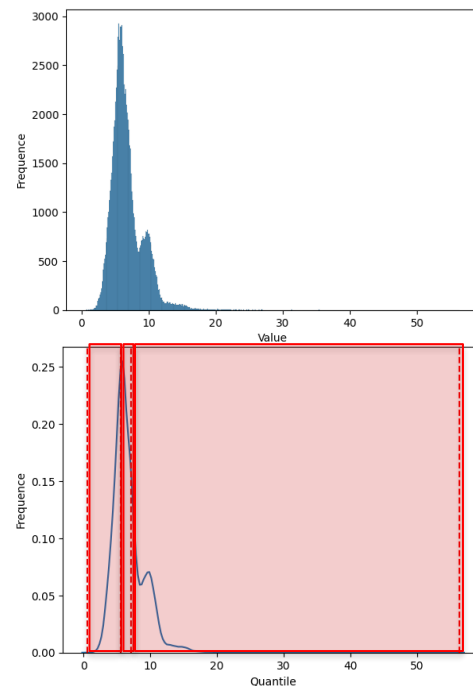
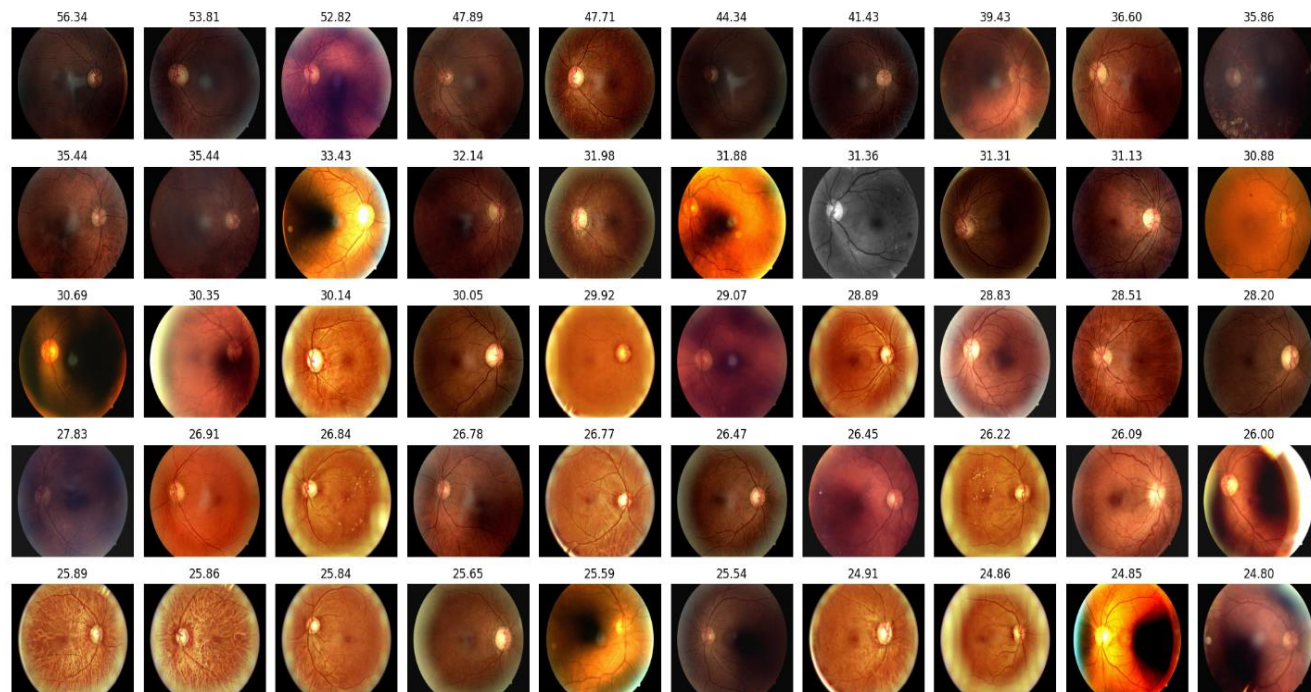
Dataset-level IQA: Brightness



pyMDMA

Input Data Auditing Metrics

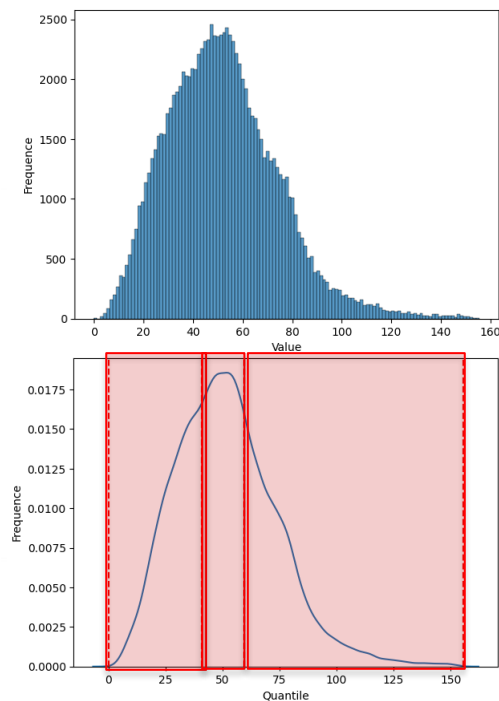
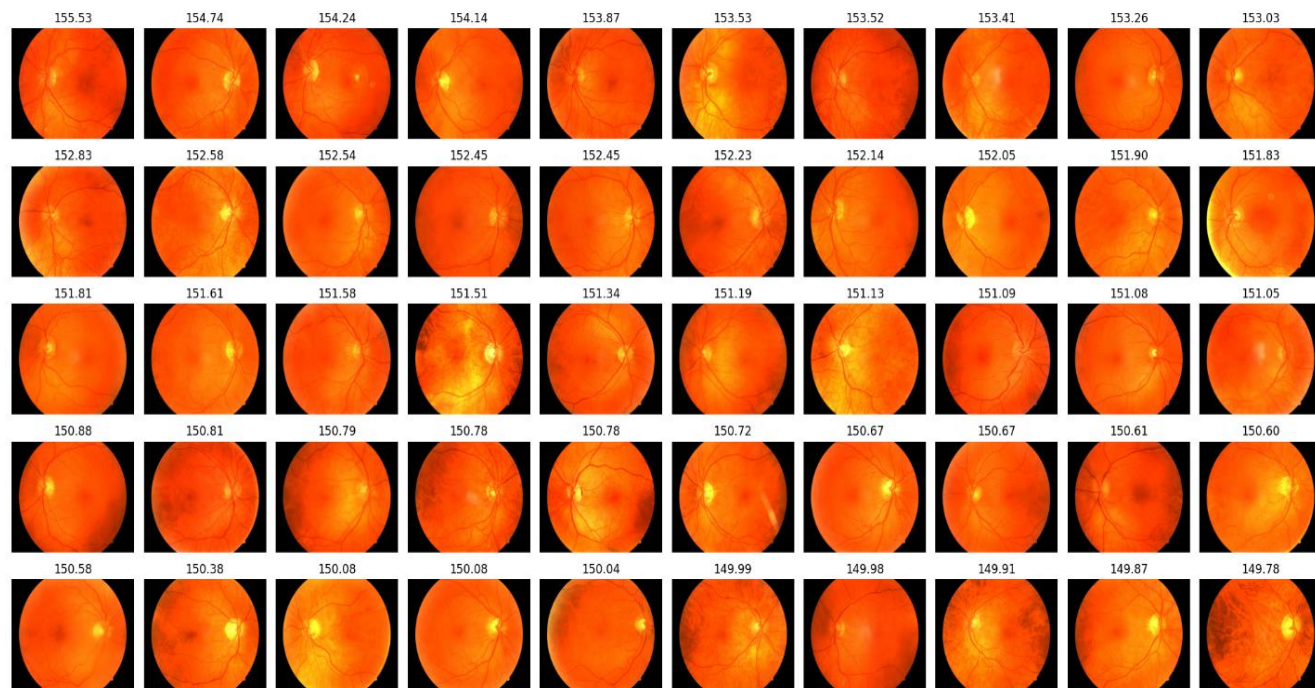
Dataset-level IQA: Sharpness (Tenengrad)



pyMDMA

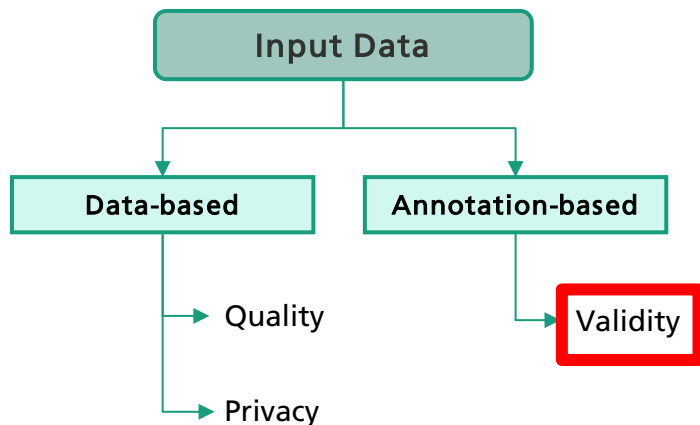
Input Data Auditing Metrics

Dataset-level IQA: Colorfulness



pyMDMA

Input Data Auditing Metrics



Completeness

- E.g. All samples have (at least) one assigned label?

Correctness

- E.g. Invalid categories not comprised in the list of valid labels?

Uniqueness

- E.g. Duplicated annotations?

AI Sym4Med Data Auditor

AI Sym4Med Functionality

Selection of data auditing goals for:

- New uploaded dataset
- A subset of a dataset on the CBDB
- A combined dataset (e.g. real + synthetic data)

Save data to Workspace

✓ **Project**
Epilepsy research

✓ **Dataset details**
Dataset name

3 Audit data quality

4 Data quality report

Step 3 of 4

Audit data quality

As a subset, the data distribution may differ from the source dataset. The system will check it against the predefined data quality dimensions below. You can choose to skip the audit or reject certain dimensions.

Review predefined data quality dimensions

Dimensions	Consider 4 of 4
Completeness ⓘ	☒
Consistency ⓘ	☒
Uniformity ⓘ	☒
Uniqueness ⓘ	☒

Cancel

Previous

Skip Audit and save subset

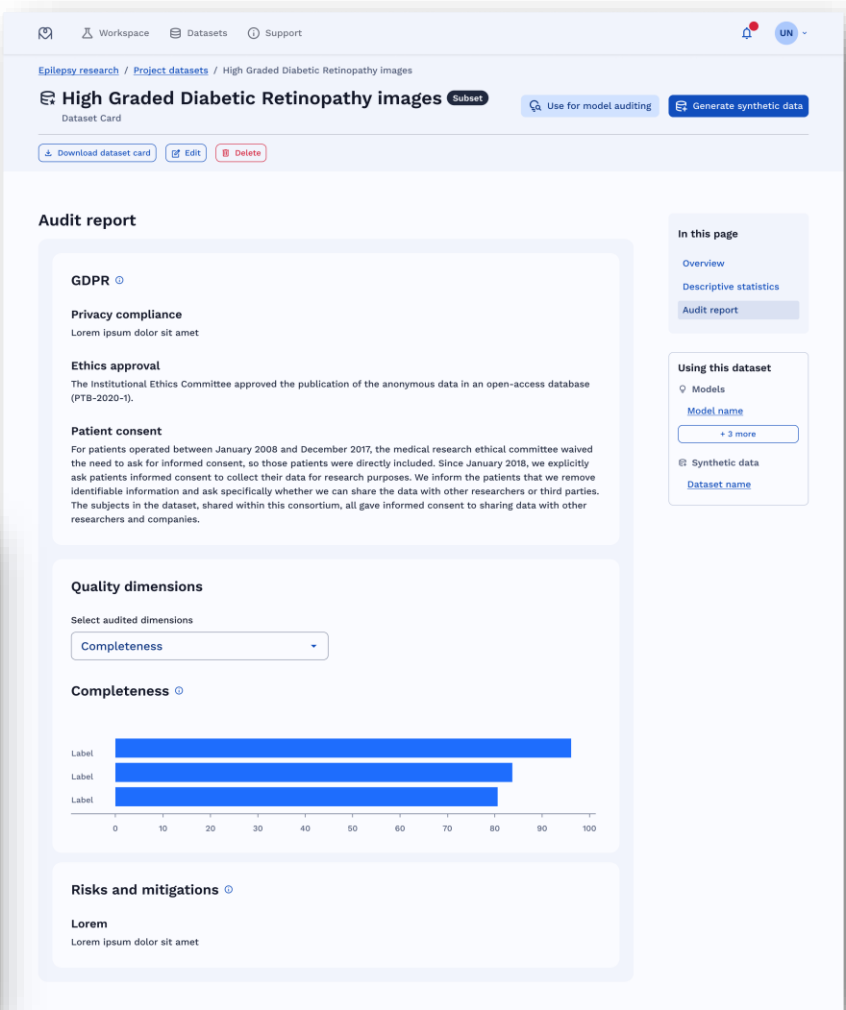
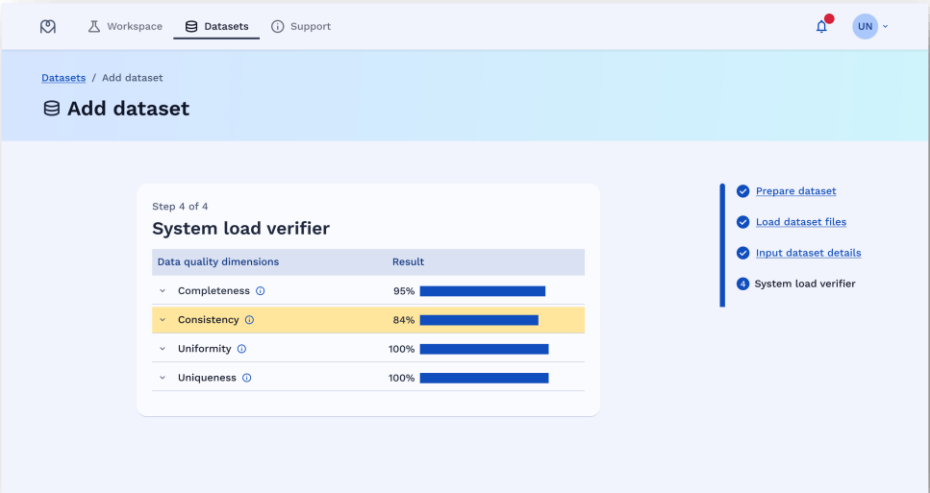
Audit subset

AI Sym4Med Data Auditor

AI Sym4Med Functionality

Dataset Card Creation

- Includes a Data Audit Report



pyMDMA

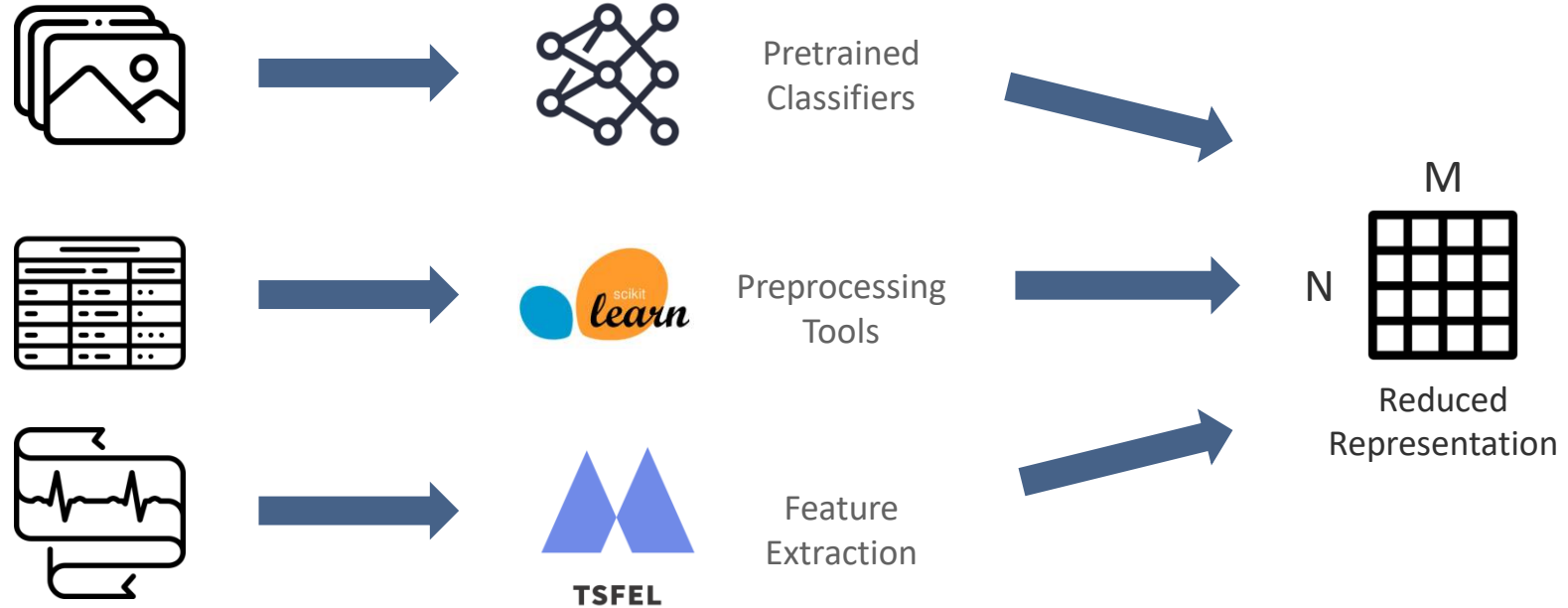
Synthetic Data Auditing Metrics

Modality	Metric	Output	Ref. Type	Category	Group	Goal	Reference
<i>General</i>	ImprovedPrecision	Dataset, Instance	Dataset	Feature-based	Quality	Fidelity	Kynkäänniemi et al. (2019)
	ImprovedRecall	Dataset, Instance	Dataset	Feature-based	Quality	Diversity	Kynkäänniemi et al. (2019)
	Authenticity	Dataset, Instance	Dataset	Feature-based	Privacy	Authenticity	Alaa et al. (2022)
	Density	Dataset, Instance	Dataset	Feature-based	Quality	Fidelity	Naeem et al. (2020)
	Coverage	Dataset, Instance	Dataset	Feature-based	Quality	Diversity	Naeem et al. (2020)
	FrechetDistance	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Yu et al. (2021)
	PrecisionRecallDistribution	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Sajjadi et al. (2018)
	GeometryScore	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Khrulkov and Oseledets (2018)
	MultiscaleIntrinsicDistance	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Tsitsulin et al. (2020)
	UtilityScore	Dataset	Dataset	Data-based	Utility	Performance	—
	WassersteinDistance	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Flamary et al. (2021)
	MMD	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Gretton et al. (2008)
	CosineSimilarity	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Manning et al. (2008)
<i>Image</i>	GIQA	Dataset, Instance	Dataset	Feature-based	Quality	Fidelity, Diversity	Gu et al. (2020)
	CMMD	Dataset	Dataset	Feature-based	Quality	Fidelity, Diversity	Jayasumana et al. (2024)
<i>Time Series</i>	DTW	Dataset, Instance	Dataset	Data-based	Quality	Fidelity, Diversity	Salvador and Chan (2007)
	CrossCorrelation	Dataset, Instance	Dataset	Data-based	Quality	Fidelity, Diversity	Proakis and Manolakis (1996)
<i>Tabular</i>	StatisticalSimScore	Dataset	Dataset	Data-based	Quality	Fidelity	Yang et al. (2024)
	StatisticalDivergenceScore	Dataset	Dataset	Data-based	Quality	Fidelity	Fonseca and Bacao (2023)
	CoherenceScore	Dataset	Dataset	Data-based	Quality	Fidelity	Yang et al. (2024)
	NNDPrivacy	Dataset	Dataset	Feature-based	Privacy	Authenticity	Liu et al. (2024)
	DCRPrivacy	Dataset	Dataset	Feature-based	Privacy	Authenticity	Liu et al. (2024)

Synthesis Validation

Feature-based (Data Embedding)

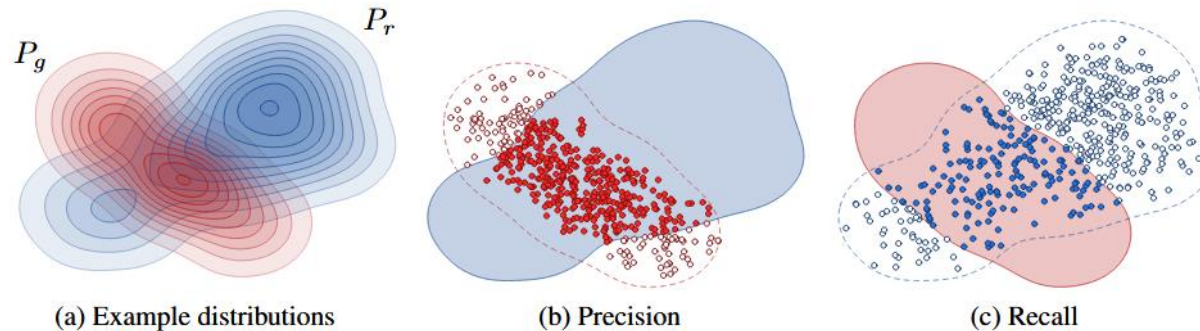
- Metrics computed on raw data are **inefficient**
- Data is compressed into **lower dimensions** for **tractable** computation



Synthesis Validation

Fidelity and Diversity

- **Fidelity** measures the **realism** of the synthetic dataset in relation to the real one
- **Diversity** measures the **coverage** of the real dataset by the synthetic one
- Most widely used metrics for synthetic evaluation measure both qualities
- Metrics such as Improved Precision and Recall^[1] decouple these concepts:



Improved Precision and Recall definitions on real (P_r) and synthetic (P_g) distributions. Reprinted from [1].

Synthesis Validation

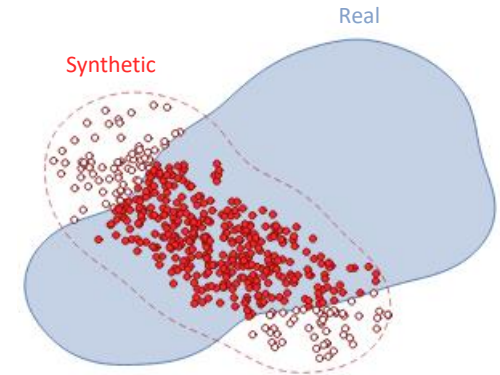
Improved Precision (Fidelity)

```
from pymdma.image.measures.synthesis_val import ImprovedPrecision
```

```
ip = ImprovedPrecision(k=5) # Fidelity metric  
# compute on NxM matrices from real and synthetic datasets  
ip_result = ip.compute(real_features, synth_features)  
precision_dataset, precision_instance = ip_result.value
```

Dataset Fidelity: 0.83

Instance Fidelity: [0, 1, 1, ..., 0, 1, 1]



Synthesis Validation

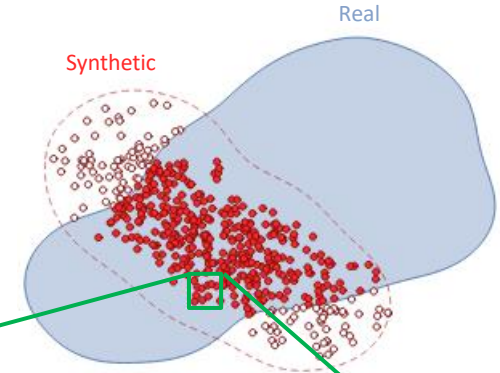
Improved Precision (Fidelity)

```
from pymdma.image.measures.synthesis_val import ImprovedPrecision
```

```
ip = ImprovedPrecision(k=5) # Fidelity metric  
# compute on NxM matrices from real and synthetic datasets  
ip_result = ip.compute(real_features, synth_features)  
precision_dataset, precision_instance = ip_result.value
```

Dataset Fidelity: 0.83

Instance Fidelity: [0, 1, 1, ..., 0, 1, 1]



Precision = 1



Synthesis Validation

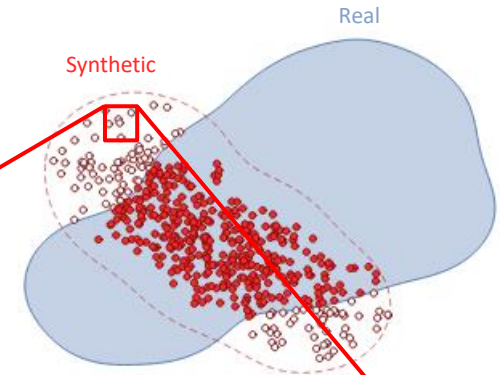
Improved Precision (Fidelity)

```
from pymdma.image.measures.synthesis_val import ImprovedPrecision
```

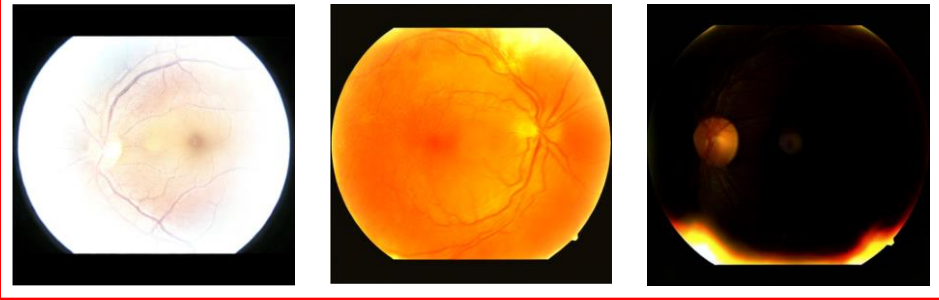
```
ip = ImprovedPrecision(k=5) # Fidelity metric  
# compute on NxM matrices from real and synthetic datasets  
ip_result = ip.compute(real_features, synth_features)  
precision_dataset, precision_instance = ip_result.value
```

Dataset Fidelity: 0.83

Instance Fidelity: [0, 1, 1, ..., 0, 1, 1]



Precision = 0



Synthesis Validation

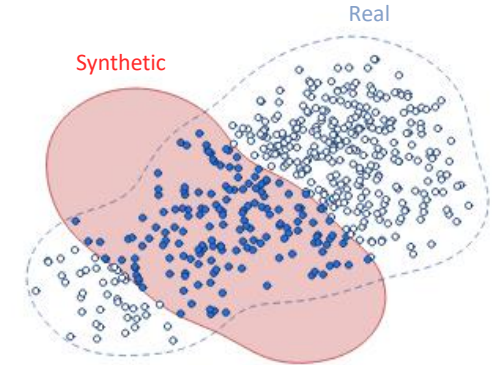
Improved Recall (Diversity)

```
from pymdma.image.measures.synthesis_val import ImprovedRecall
```

```
ir = ImprovedRecall(k=5) # Diversity metric  
# compute on NxM matrices from real and synthetic datasets  
ir_result = ir.compute(real_features, synth_features)  
recall_dataset, recall_instance = ir_result.value
```

Dataset Diversity: 0.62

Instance Diversity: [0, 2, 3, ..., 0, 9, 1]



Synthesis Validation

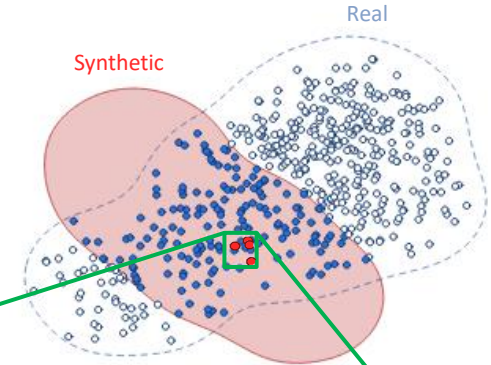
Improved Recall (Diversity)

```
from pymdma.image.measures.synthesis_val import ImprovedRecall

ir = ImprovedRecall(k=5) # Diversity metric
# compute on NxM matrices from real and synthetic datasets
ir_result = ir.compute(real_features, synth_features)
recall_dataset, recall_instance = ir_result.value
```

Dataset Diversity: 0.62

Instance Diversity: [0, 2, 3, ..., 0, 9, 1]



Recall > 0



Synthesis Validation

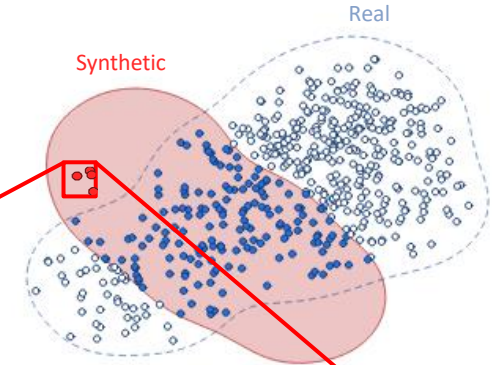
Improved Recall (Diversity)

```
from pymdma.image.measures.synthesis_val import ImprovedPrecision
```

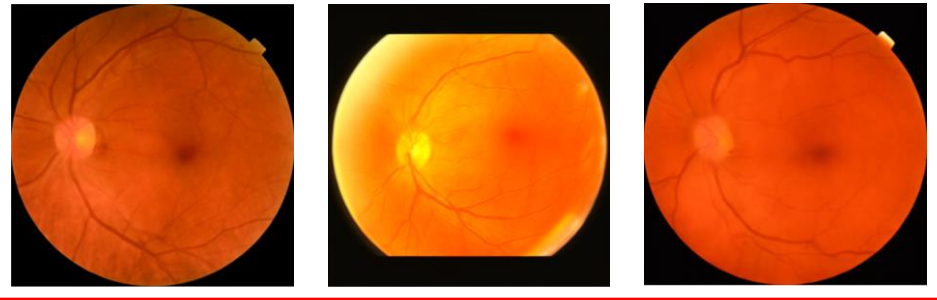
```
ir = ImprovedRecall(k=5) # Diversity metric  
# compute on NxM matrices from real and synthetic datasets  
ir_result = ir.compute(real_features, synth_features)  
recall_dataset, recall_instance = ir_result.value
```

Dataset Diversity: 0.62

Instance Diversity: [0, 2, 3, ..., 0, 9, 1]



Recall = 0





Check out
pyMDMA Library

Install and Contribute



Install with pip:

```
pip install pymdma
```

Don't forget the extras:

```
pip install pymdma[image] # image dependencies
pip install pymdma[tabular] # tabular dependencies
pip install pymdma[time_series] # time series dependencies
pip install pymdma[all] # dependencies for all modalities
```



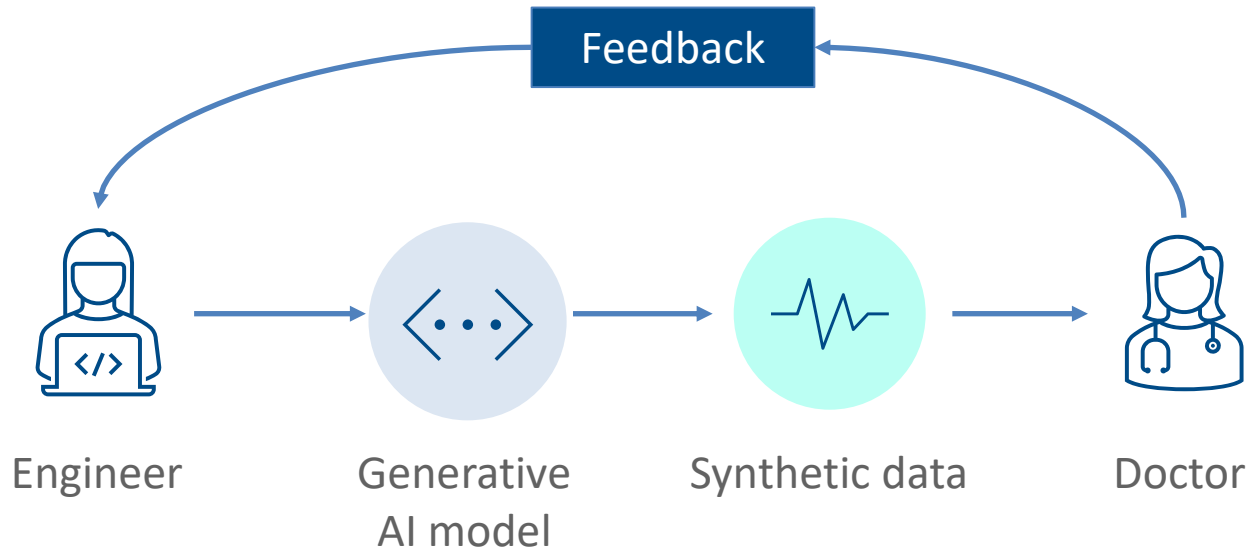
Leave a star on GitHub

<https://github.com/fraunhoferportugal/pymdma.git>

Feedback is always welcome! 😊

Qualitative Synthetic Data Auditing

Doctor-In-The-Loop Approach



AI Sym4Med Data Auditor

Doctor-In-The-Loop Functionality

Synthetic retinal fundus images dataset For review

- 1 **Generation Criteria**
- 2 Global analysis
- 3 Visual analysis
- 4 Overall review

About the input

 **Consider that**

Limitations of the real dataset can be propagated into the synthetic data, leading to biases and shortcomings for underrepresented cases.

Real dataset used to train generative model

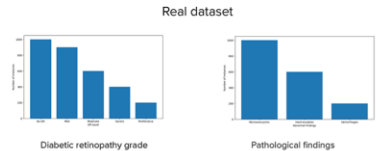
Real dataset Retinal fundus images dataset

Summary

- Target disease(s): Diabetic retinopathy
- Acquisition modality: Retinal fundus image
- Number of records (rows): 3050
- Annotation classes: Diabetic retinopathy grade - No Diabetic retinopathy, Mild, Moderate, Severe and Proliferative + Pathological findings - Microaneurysms, Hemorrhages, Hard exudates, and Neovascularization
- Human annotation by 3 ophthalmologists

Generation purpose

Balance train set class distributions to train predictive ML model for Diabetic retinopathy



Generative model used

Generative model name Model name

Description 2 lines maximum

Generation options available The generative model used provides control:

- 1 - Image characteristics **specific** of the target clinical domain
 - Disease grade (disease grade of each image to be generated)
 - Morphological findings (type, location, size and/or shape of the controllable morphological findings of each image to be generated)
 - Pathological findings (type, location, size and/or shape of each controllable pathological component to be generated)
- 2 - Image quality characteristics **agnostic** to the target clinical domain
 - Brightness (brightness levels for each image to be generated)
 - Sharpness (sharpness levels for each image to be generated)
 - Focus (focus levels for each image to be generated)
- 3 - **Representativeness between real and synthetic** data in terms of:
 - Location of pathological findings by Diabetic retinopathy grade
 - Size of pathological findings
 - Co-occurrence of pathological findings by Diabetic retinopathy grade

Generation criteria

 **Consider that**

The generative model uses the values defined for generation - as specified in the *Synthetic dataset* columns. However, the output may vary.

1 - Image characteristics **specific** of the target clinical domain

Disease grade - Synthetic data total count Give feedback

	Synthetic dataset	Real dataset	Synthetic + Real dataset
No DR	0	1000	1000
Mild	100	900	1000
Moderate	400	600	1000
Severe	650	350	1000
Proliferative	800	200	1000

AI Sym4Med Data Auditor

Doctor-In-The-Loop Functionality

Synthetic retinal fundus images dataset For review

- 1 Generation Criteria
- 2 **Global analysis**
- 3 Visual analysis
- 4 Overall review

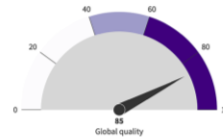
About the output

*Based on statistical metrics computed by the system

Global Quality ?

How well synthetic data captures properties of real data

Score
85



Fidelity ?

How similar the synthetic data is to the real data.

Score
72

Privacy ?

How well synthetic data prevents re-identification.

Score
42

Diversity ?

How well the synthetic data captures the variability of the real data.

Score
88

Utility ?

How well the synthetic data performs when trained on ML classification models.

Score
68

① **Inter-dependent scores.** Higher fidelity means the data is very realistic but might not protect privacy well. Low fidelity means the data isn't as realistic but may be better for privacy.

AI Sym4Med Data Auditor

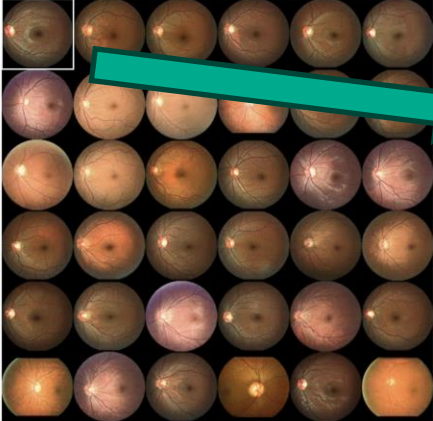
Doctor-In-The-Loop Functionality

Synthetic retinal fundus images dataset For review

1 Generation Criteria 2 Global analysis 3 Visual analysis 4 Overall review

Filter synthetic images by (Diabetic retinopathy grade) Moderate Sort by Uncertainty

Moderate (40) Show 36 per frame

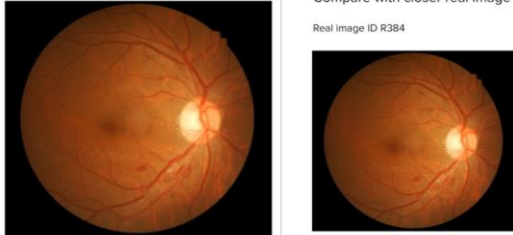


Previous 1 of 2 frames Next

Synthetic image ID S384 Accept Reject

Compare with closer real image

Real image ID R384



Diabetic retinopathy grade: **Moderate**
Pathological findings: **Microaneurysm, Hemorrhage**

Reject synthetic image ID S384

Problems found

☐ Unrealistic morphological features

- ☐ Blood vessels
- ☐ Macula
- ☐ Fovea
- ☐ Optic disk
- ☐ Other

☐ Unrealistic pathological findings

- ☐ Location
- ☐ Co-occurrence
- ☐ Size
- ☐ Other

☐ Inadequate image quality characteristics, related with:

- ☐ Brightness
- ☐ Contrast
- ☐ Focus
- ☐ Aspect ratio
- ☐ Other

☐ No correlation between diagnostic category and image. It should be:

- ☐ No DR
- ☐ Mild
- ☐ Moderate
- ☐ Severe
- ☐ Proliferative
- ☐ Other

☐ Other

Add comment

Submit review

AI Sym4Med Modules

 AI Sym4Med

 AI Sym4Med Use Cases

 Support

Login

Register

AI SYM4MED

To improve the healthcare digital ecosystem.

 **Dataset explorer**

Collect, aggregate and analyse health data for secure access, exploration, and synthesis.

[View Demo](#)

 **Synthetic Data Generator**

Enhance experimentation capacities via the expansion of datasets with reliable synthetic data. This tool helps overcome privacy concerns and data scarcity by producing high-quality, artificial data that mirrors real-world medical scenarios.

[View Demo](#)

 **Data Auditor**

Ensure data quality, trustworthiness and privacy.

[View Demo](#)

 **Model Auditor**

Analyse and characterise AI models in terms of model performance, model bias, model limitations and compliance.

[View Demo](#)

AI SYM4MED

Terms & Policies

Join Community of practice

Give Feedback

Model Auditing: analysis pillars

A model auditing repository that analyses different model properties under four main pillars

Goal



Robustness



Fairness



Privacy



Explainability

1

Supported data modalities:

- Tabular
- Image
- Time-series

2

Supported models:

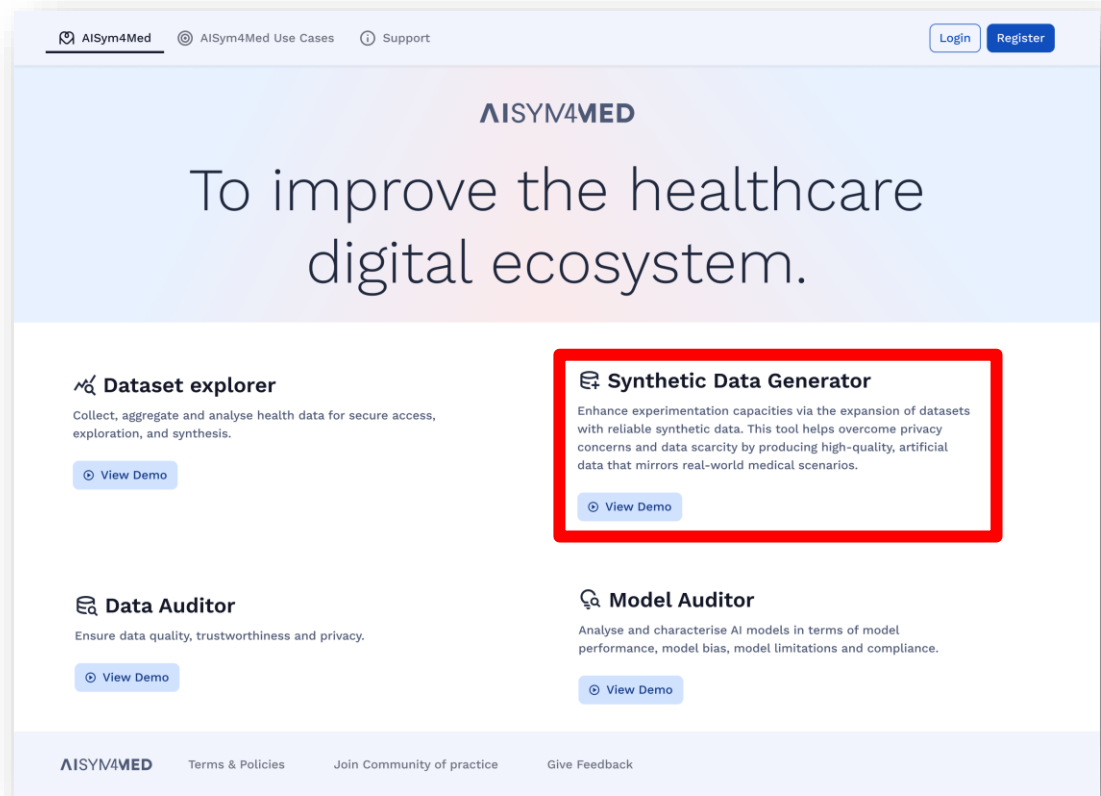
- Deep Neural Networks
- Sci-kit learn ML models
- Boosting models

3

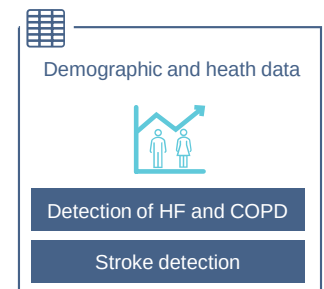
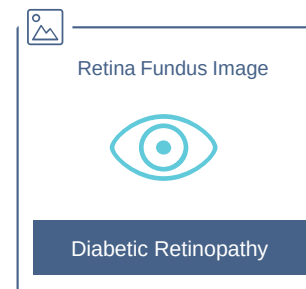
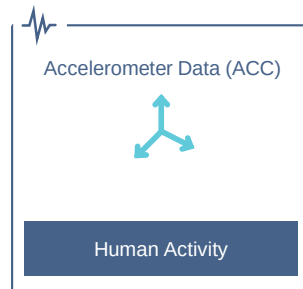
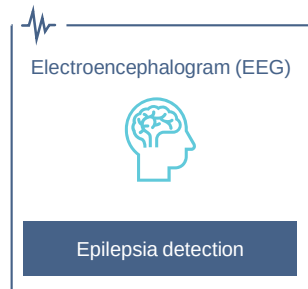
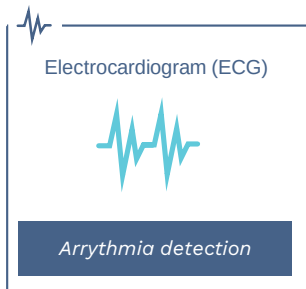
Supported model tasks:

- Classification
- Regression

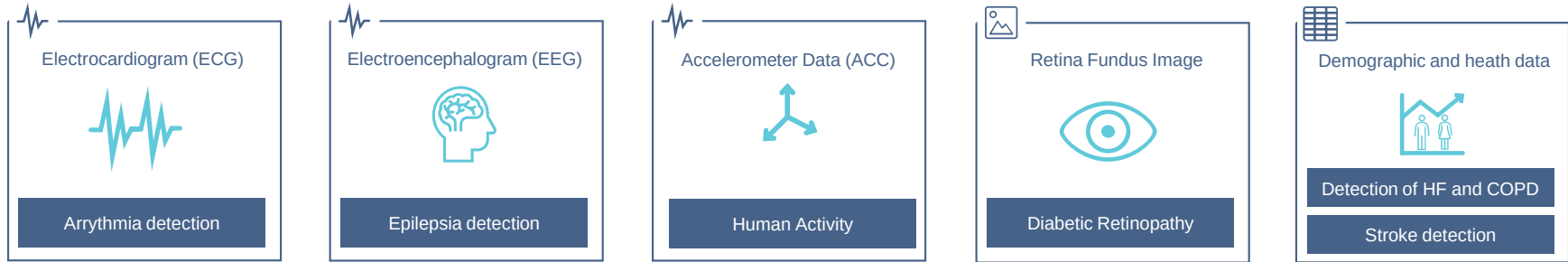
AI Sym4Med Modules



Synthetic Data Generation

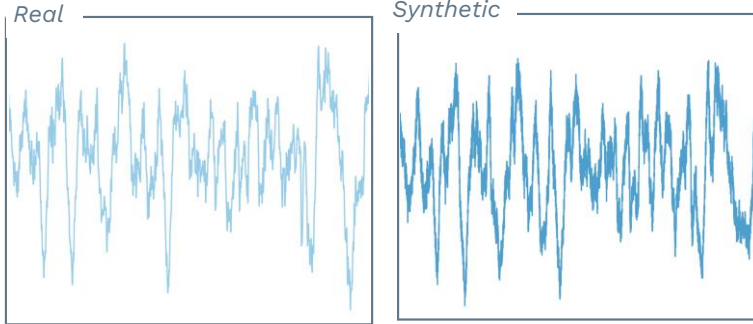


Synthetic Data Generation: Preliminary Results

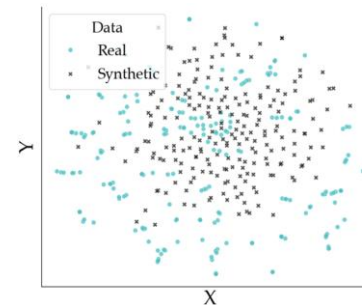


Generative Adversarial Network (GAN)

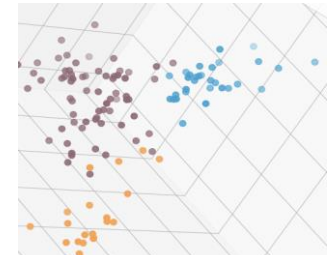
Fidelity^{*1}



Diversity^{*2}



Utility^{*3}



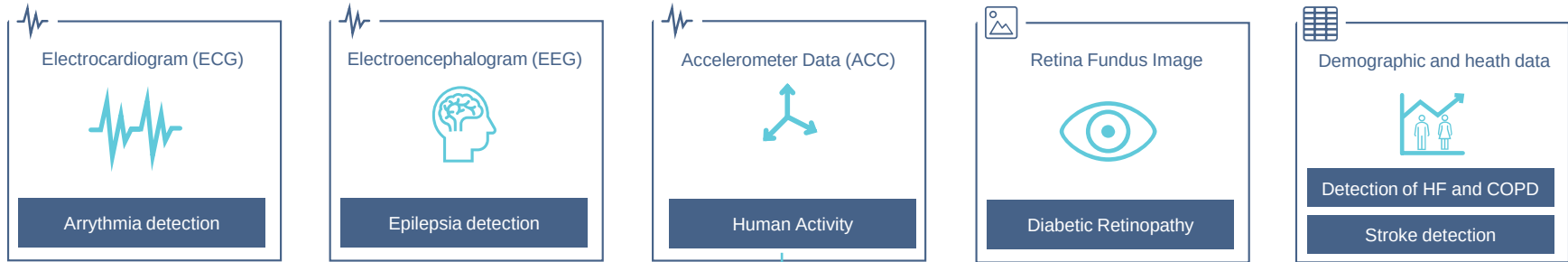
Real: 87 %
Real + Synthetic: 93 %

^{*1} *Fidelity*: Realism of the synthetic samples produced

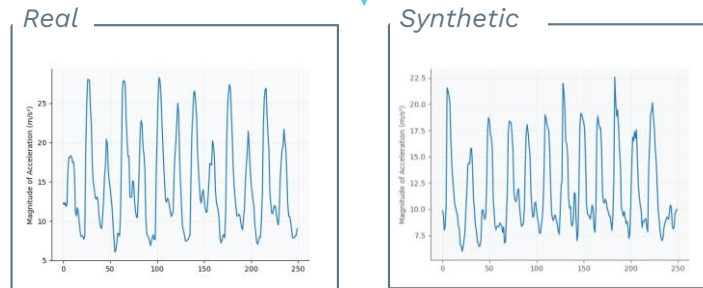
^{*2} *Diversity*: Variety of the synthetic samples produced

^{*3} *Utility*: Usefulness of the synthetic data for a specific task

Synthetic Data Generation: Preliminary Results



Diffusion Model



ACC signals example

Synthesis Quality

Fidelity *1	Diversity *2
66 %	70 %

Synthesis Utility *2

Train on:	Test on Real
Real	81.21%
Synthetic	79.73%
Real + Synthetic	81.55%

Downstream task

Evaluation metric: F1-Score

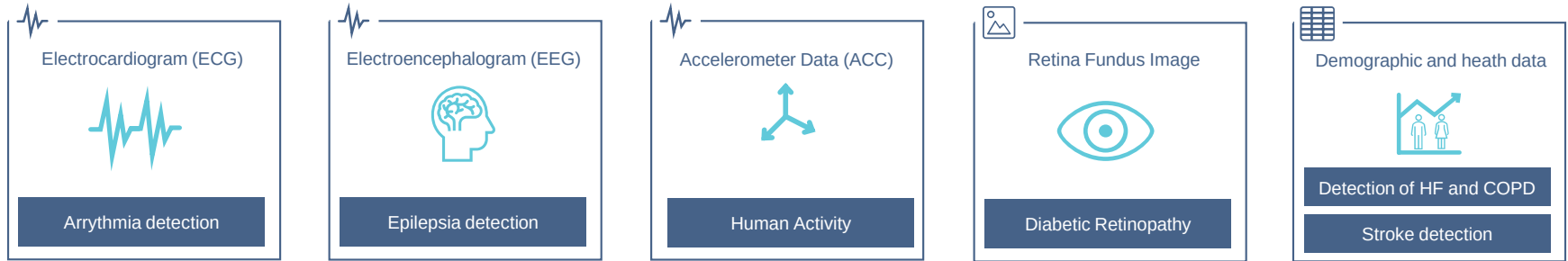
Classification model: SVM

*1 **Fidelity**: Realism of the synthetic samples produced

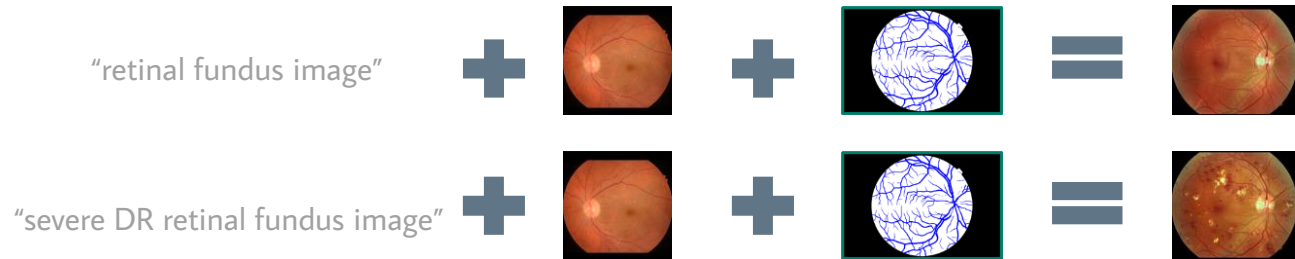
*2 **Diversity**: Variety of the synthetic samples produced

*3 **Utility**: Usefulness of the synthetic data for a specific task

Synthetic Data Generation: Preliminary Results



SD20 ZSNR + ControlNet
Model



Use-Case 1

Diabetic Retinopathy Detection and Prediction

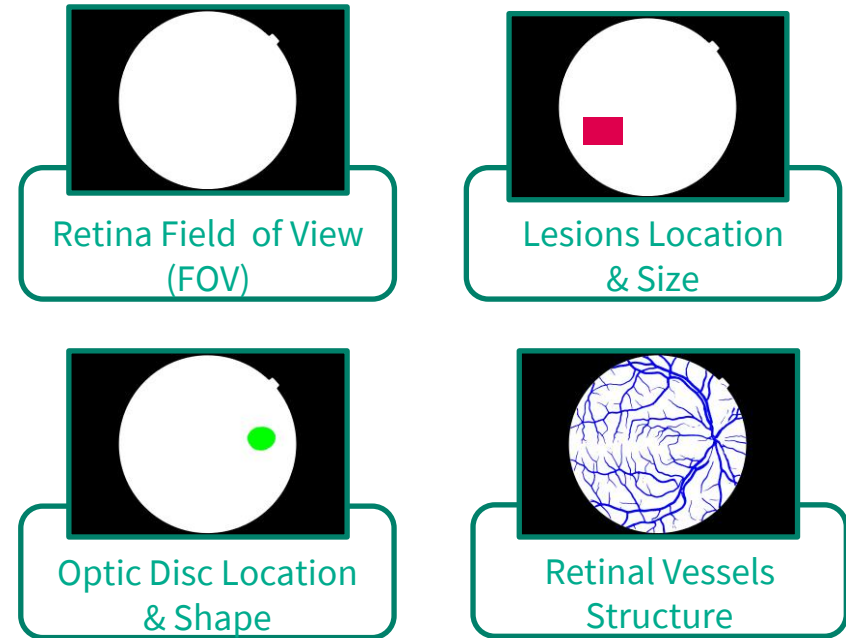
Objectives

- **Task 1:** Predict diabetic retinopathy from tabular data registered in Health Information Systems (HIS)
- **Task 2:** Generate retinal fundus images with diabetic retinopathy

Pilot Opportunities

- Data discovery, exploration, evaluation and splitting in federated nodes
- Federated tabular and image-based model training
- Development of conditional image generation
- Evaluation of synthetic data and Doctor-in-the-loop approach
- Model auditing and generalization ability

Conditioning Image Masks



Use-Case 2

Retrospective extraction of clinical knowledge

Objectives

- Convert data produced during clinical care, into usable data to:
 - Search and analyze existing data for identification of candidates for cohort selection;
 - Extract scientific knowledge about the diagnosis, treatment and prognosis of diseases.

Opportunities

- De-identification of unstructured data
- Data structuring and standardization
- Federated based data discovery, exploration and complex searches
- Evaluation of federated data, review of data profiles and statistical information of the data



Clinical Information

Data analysis

Select patients for trials



Use-Case 3

Advanced Parkinson Disease (APD) Prediction

Objectives

- To predict APD from retrospective clinical information, and from prospective trials

Opportunities

- Data structuring and standardization for ML purposes
- Augmentation with synthetic data
- Comprehensive model auditing



Use-Case 4

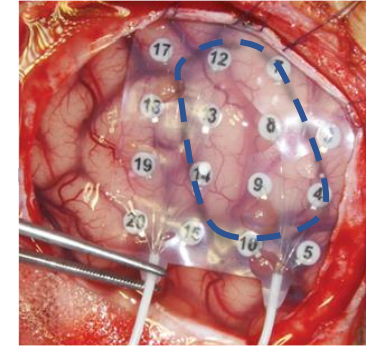
Epilepsy Surgery Planning

Objectives

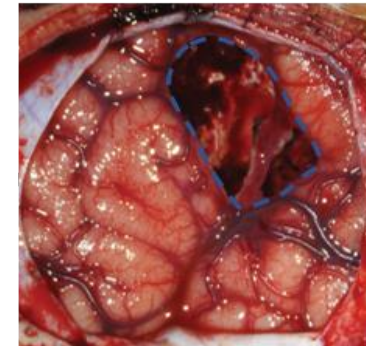
- *Identify epileptic tissue to be removed by surgery from intraoperative electrocorticography (ioECoG) data.*

Opportunities

- *Data structuring and standardization for ML purposes*
- *Augmentation with synthetic data*
- *Comprehensive model auditing*



*Intraoperative
electrocorticography*



Resection

Use-Case 5

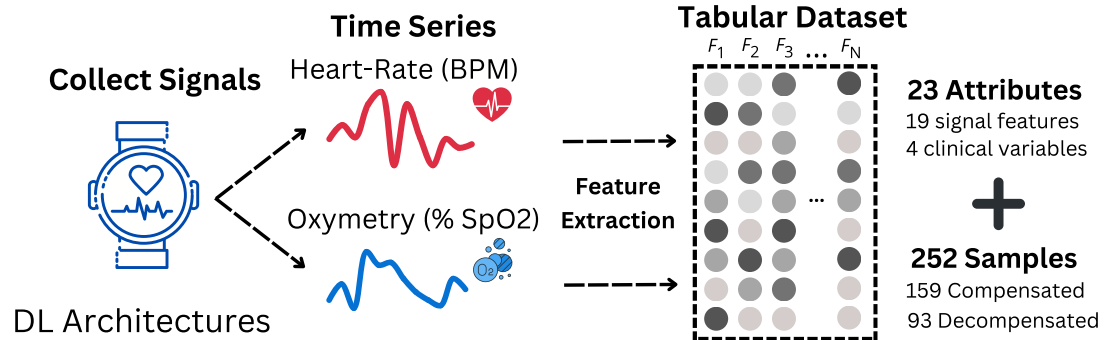
Chronic heart and respiratory diseases monitoring

Objectives

- Detect worsening of chronic heart and respiratory diseases from patient monitoring data

Opportunities

- Utilize open data from previous research project (limited size)
- Augmentation with synthetic data
- Comprehensive model auditing



COMMUNITY OF PRACTICE

The AISym4MED Community of Practice (CoP) aims to bring together experts and entities from different fields to improve secure health data processing and the robustness of medical AI solutions.

YOU WILL BE INVITED TO:

- ✓ Information-sharing activities
- ✓ Workshops
- ✓ Meetings
- ✓ Presentation events



AISYM4MED

MAIN GOALS



Shape the global conversation on **breaking health data silos** for the development of robust medical AI solutions



Build expertise on **AI model assessment** for medical applications



Foster the development of synthetic health data generation



Advance the knowledge on **effective evaluation and usage of synthetic health data**

ACTIVITIES PER TARGET AUDIENCE



AI developers/ Medical AI companies

1. Feedback on AISym4MED platform
2. Hackathons



Scientists, academics or research institutions

1. Workshops
2. Knowledge sharing
3. Joint publications
4. Internships



Clinicians, Clinical researchers, Healthcare institutions

1. Clinical evaluation of synthetic data
2. Doctor-in-the-loop
3. Bring your Use Cases



Interested in health data

1. Ethical and legal aspects related to health data and AI
2. Policy recommendations



Co-funded by
the European Union

Funded by the European Union under the Horizon Europe Framework Programme Grant Agreement N°: 101095387. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Commission. Neither the European Union nor the European Health and Digital Executive Agency can be held responsible for them.

Project funded by



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research EAER
State Secretariat for Education,
Research and Innovation SERI



UK Research
and Innovation

COP CALENDAR 2025



JANUARY

FEBRUARY

MARCH

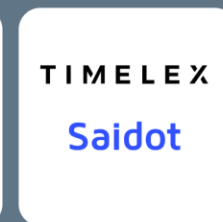
APRIL

MAY

JUNE

JULY

SEPTEMBER



ACTIVITIES



ACTIVITIES



ACTIVITIES



ACTIVITIES



ACTIVITIES



ACTIVITIES



ACTIVITIES



ACTIVITIES

28/01 Kick off webinar (Online)

27/02 Generative AI for Synthetic Health Data (Online)

27/03 Workshop on medical data integration across borders (Online)

02/04 Focus group on Synthetic Ophthalmology Image Data (Porto)
09/04 Focus group session (Ankara)

22/05 Showcase & Feedback Session for AI developers (Online)

04/06 Focus group on AI compliance (Online)

02/07 Focus group on clinical evaluation of synthetic data (Porto)

25/09 Workshop on medical AI model auditing



Co-funded by
the European Union

Funded by the European Union under the Horizon Europe Framework Programme Grant Agreement N°: 101095387. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Commission. Neither the European Union nor the European Health and Digital Executive Agency can be held responsible for them.

Project funded by



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research - EAER
State Secretariat for Education,
Research and Innovation SERI



UK Research
and Innovation

WORKING GROUPS



WG 1 Research and innovation on health data in the EU

WG 2 Effective evaluation and usage of synthetic health data

WG 3 Ethical and legal aspects of medical AI



Co-funded by
the European Union

Funded by the European Union under the Horizon Europe Framework Programme Grant Agreement N°: 101095387. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Commission. Neither the European Union nor the European Health and Digital Executive Agency can be held responsible for them.

Project funded by

 Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra
Swiss Confederation

Federal Department of Economic Affairs,
Education and Research EAER
State Secretariat for Education,
Research and Innovation SERI



UK Research
and Innovation



AISym4MED



aisym4med.eu



@AISym4MED_Project



Co-funded by the
European Union

Project funded by



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research EAER
**State Secretariat for Education,
Research and Innovation SERI**



UK Research
and Innovation

AISYM4MED

To improve the healthcare AI ecosystem

Thank you. Follow us and discover more.

Luís Rosado | E-mail: luis.rosado@fraunhofer.pt



Funded by the European Union under Grant Agreement N.210810650. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the granting authorities. Neither the European Union nor the granting authorities can be held responsible for them.



Project funded by



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research SAER
State Secretariat for Education,
Research and Innovation SERI